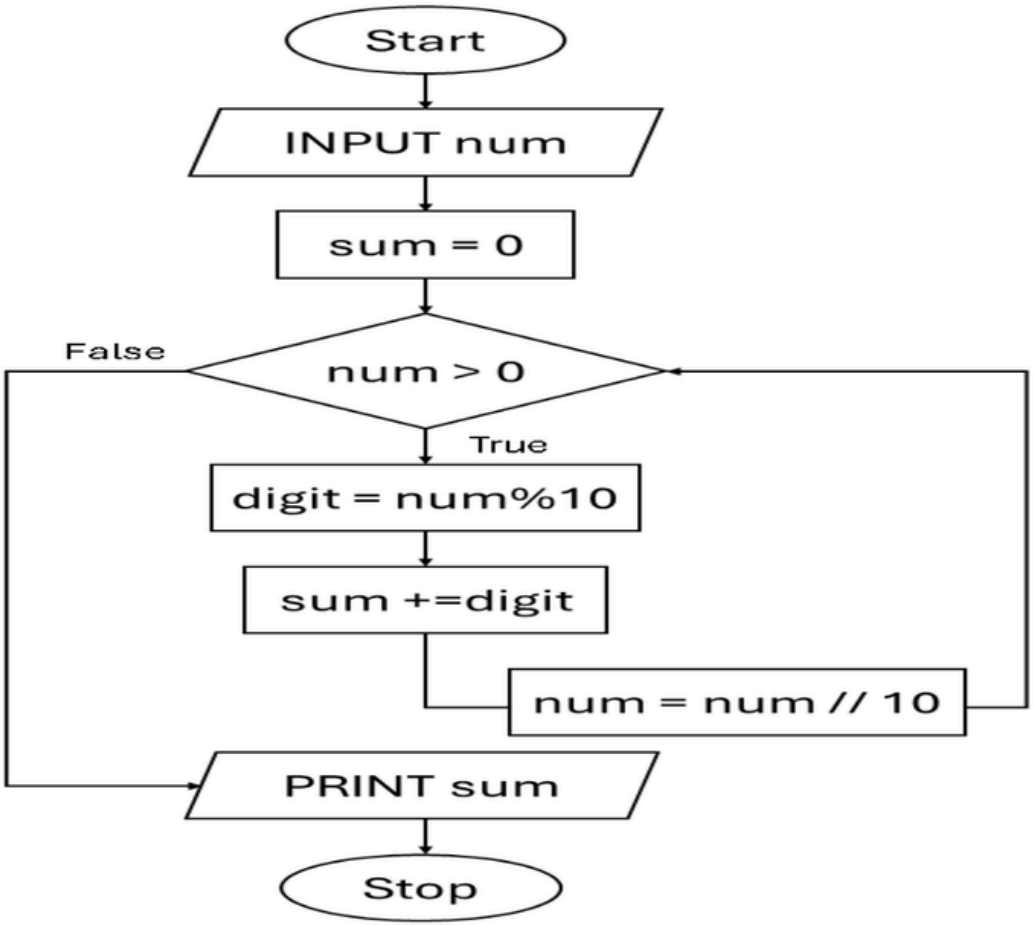
Experiment 11

11.1.1 Sum of digits

Algorithm :

Step 1 : Start
Step 2 : Input an integer and store it in num
Step 3 : Set sum to 0
Step 4 : Repeat steps 5 through 7 while num is greater than 0
Step 5 : Find the last digit using the modulo operator:
digit = num % 10
Step 6 : Add the digit to the current sum
Step 7 : Remove the last digit using integer division:
num = num // 10
Step 8 : Go back to Step 4
Step 9 : Once num becomes 0, exit the loop
Step 10 : Print the final value of sum
Step 11 : Stop

Flowchart :



11.1.1. Sum of Digits of a Number

06:20

Write a Python program that reads an integer, calculates the sum of its digits, and prints the result.

Input Format:

- The program reads an integer input with the prompt "Enter a number: "

Output Format:

- Print the sum of the digits in the format:

```
Sum of digits: <digit_sum>
```

Note:

- Some base code was provided (input and print statements). The task was to complete the missing logic.

Sample Test Cases +

Explorer

sumofdig...

Submit

```python
num = int(input("Enter a number: "))
sum=0
while num>0:
    i=num%10
    sum+=i
    num//=10
```

Debugger

Average time	Maximum time
0.011 s	0.015 s
11.00 ms	15.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1 15 ms Debug

Expected output	Actual output
Enter a number: 123	Enter a number: 123
Sum of digits: 6	Sum of digits: 6

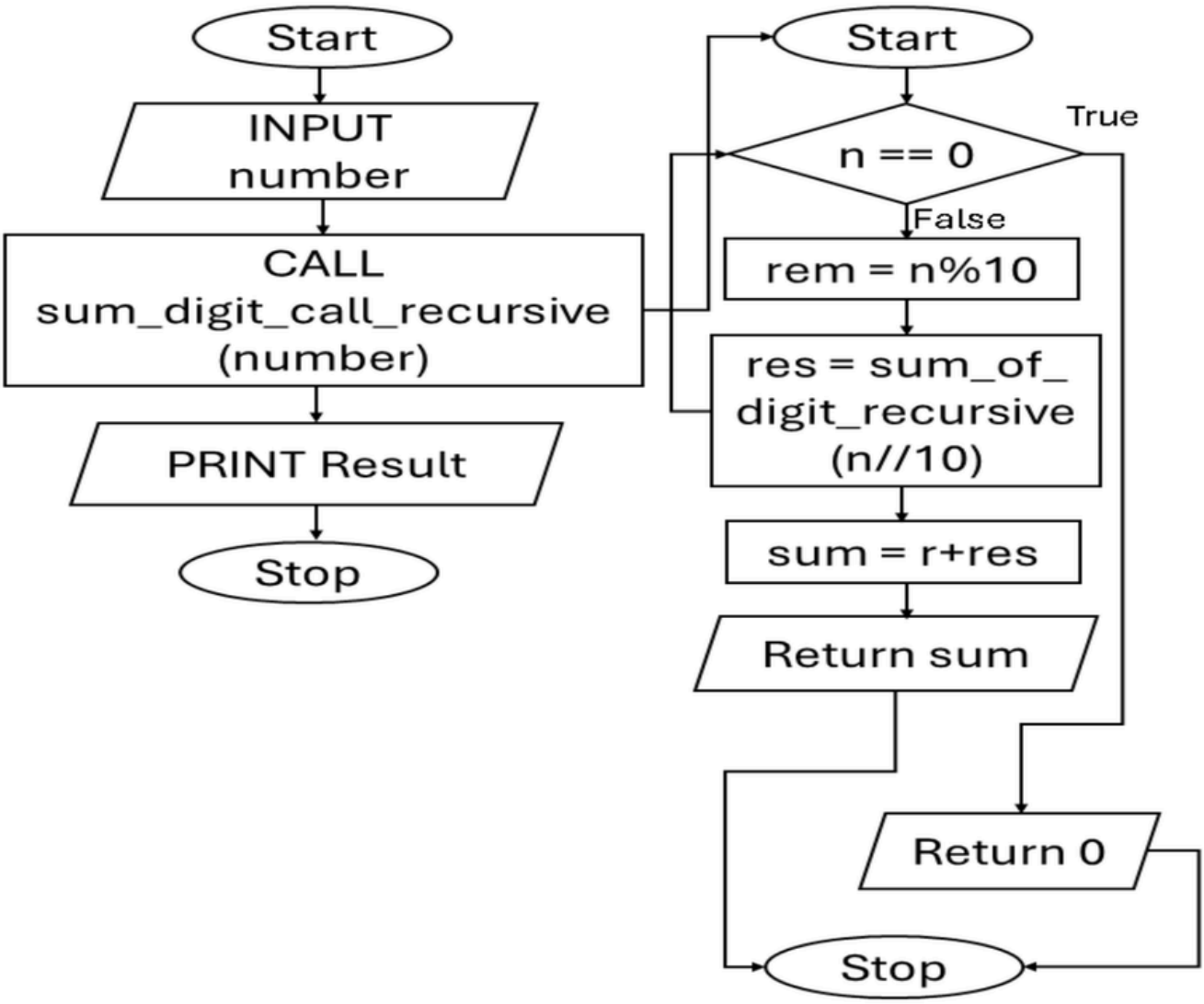
Experiment 11

11.1.2 Sum of digits using recursion

Algorithm :

Step 1: Start
Step 2: Input an integer n
Step 3: Call function sum(n)
Step 4: If n == 0, return 0
Step 5: Else compute:
    a) r = n % 10 (last digit)
    b) q = n // 10 (remaining number)
Step 6: Return r + sum(q)
Step 7: Print the result
Step 8: Stop

Flowchart :



11.1.2. Sum of Digits using Recursion

07:14

Write a recursive function **sum_of_digits_recursive(n)** that calculates the sum of the digits of a given non-negative integer n.

Input Format:

- The input consists of a single non-negative integer n.

Output Format:

- The output should be the sum of the digits of the non-negative integer n.

Constraints:

- $0 \leq n \leq 10^9$

Sample Test Cases +

Explorer

SumofDi...

Submit

```
def sum_of_digits_recursive(n):
    if n==0:
        return 0
    return (n%10)+sum_of_digits_recursive(n//10)
number=int(input())
```

Debugger

Average time	Maximum time
0.011 s	**0.016 s**
11.20 ms	16.00 ms

2 out of 2 shown test case(s) passed

3 out of 3 hidden test case(s) passed

Test case 1 16 ms Debug

Expected output	Actual output
23	23
5	5